

Module `java.base`
Package `java.util.stream`

Class Collectors

`java.lang.Object`
`java.util.stream.Collectors`

public final class **Collectors**
extends `Object`

Implementations of `Collector` that implement various useful reduction operations, such as accumulating elements into collections, summarizing elements according to various criteria, etc.

The following are examples of using the predefined collectors to perform common mutable reduction tasks:

```
// Accumulate names into a List
List<String> list = people.stream()
    .map(Person::getName)
    .collect(Collectors.toList());

// Accumulate names into a TreeSet
Set<String> set = people.stream()
    .map(Person::getName)
    .collect(Collectors.toCollection(TreeSet::new));

// Convert elements to strings and concatenate them, separated by commas
String joined = things.stream()
    .map(Object::toString)
    .collect(Collectors.joining(", "));

// Compute sum of salaries of employee
int total = employees.stream()
    .collect(Collectors.summingInt(Employee::getSalary));

// Group employees by department
Map<Department, List<Employee>> byDept = employees.stream()
    .collect(Collectors.groupingBy(Employee::getDepartment));

// Compute sum of salaries by department
Map<Department, Integer> totalByDept = employees.stream()
    .collect(Collectors.groupingBy(Employee::getDepartment,
                                   Collectors.summingInt(Employee::getSalary)));

// Partition students into passing and failing
Map<Boolean, List<Student>> passingFailing = students.stream()
    .collect(Collectors.partitioningBy(s -> s.getGrade() >= PASS_THRESHOLD));
```

Since:
1.8

Method Summary

All Methods	Static Methods	Concrete Methods
Modifier and Type		
static <T> Collector <T,?,Double>		
	Method	Description
	averagingDouble (ToDoubleFunction <? super T> mapper)	Returns a Collector that produces the arithmetic mean of a double-valued function applied to the input elements.
	averagingInt (ToIntFunction <? super T> mapper)	Returns a Collector that produces the arithmetic mean of an integer-valued function applied to the input elements.

OVERVIEW MODULE PACKAGE CLASS USE TREE DEPRECATED INDEX HELP

Java SE 10 & JDK 10

PREV CLASS NEXT CLASS FRAMES NO FRAMES ALL CLASSES

SEARCH:

SUMMARY: NESTED | FIELD | CONSTR | METHOD DETAIL: FIELD | CONSTR | METHOD

static <T,A,R,RR> Collector <T,A,RR>	collectingAndThen (Collector <T,A,R> downstream, Function <R,RR> finisher)	Adapts a Collector to perform an additional finishing transformation.
static <T> Collector <T,?,Long>	counting ()	Returns a Collector accepting elements of type T that counts the number of input elements.
static <T,A,R> Collector <T,?,R>	filtering (Predicate <? super T> predicate, Collector <? super T,A,R> downstream)	Adapts a Collector to one accepting elements of the same type T by applying the predicate to each input element and only accumulating if the predicate returns true.
static <T,U,A,R> Collector <T,?,R>	flatMapMapping (Function <? super T,? extends Stream <? extends U>> mapper, Collector <? super U,A,R> downstream)	Adapts a Collector accepting elements of type U to one accepting elements of type T by applying a flat mapping function to each input element before accumulation.
static <T,K> Collector <T,?, Map <K, List <T>>>	groupingBy (Function <? super T,? extends K> classifier)	Returns a Collector implementing a "group by" operation on input elements of type T, grouping elements according to a classification function, and returning the results in a Map.
static <T,K,D,A,M extends Map <K,D>> Collector <T,?,M>	groupingBy (Function <? super T,? extends K> classifier, Supplier <M> mapFactory, Collector <? super T,A,D> downstream)	Returns a Collector implementing a cascaded "group by" operation on input elements of type T, grouping elements according to a classification function, and then performing a reduction operation on the values associated with a given key using the specified downstream Collector.
static <T,K,A,D> Collector <T,?, Map <K,D>>	groupingBy (Function <? super T,? extends K> classifier, Collector <? super T,A,D> downstream)	Returns a Collector implementing a cascaded "group by" operation on input elements of type T, grouping elements according to a classification function, and then performing a reduction operation on the values associated with a given key using the specified downstream Collector.
static <T,K> Collector <T,?, ConcurrentMap <K, List <T>>>	groupingByConcurrent (Function <? super T,? extends K> classifier)	Returns a concurrent Collector implementing a "group by" operation on input elements of type T, grouping elements according to a classification function.
static <T,K,A,D,M extends ConcurrentMap <K,D>> Collector <T,?,M>	groupingByConcurrent (Function <? super T,? extends K> classifier, Supplier <M> mapFactory, Collector <? super T,A,D> downstream)	Returns a concurrent Collector implementing a cascaded "group by" operation on input elements of type T, grouping elements according to a classification function, and then performing a reduction operation on the values associated with a given key

OVERVIEW				MODULE	PACKAGE	CLASS	USE	TREE	DEPRECATED	INDEX	HELP	Java SE 10 & JDK 10	
PREV CLASS		NEXT CLASS		FRAMES	NO FRAMES	ALL CLASSES		SEARCH:					
SUMMARY: NESTED FIELD CONSTR METHOD				DETAIL: FIELD CONSTR METHOD									
				super T,A,D> downstream)				cascaded "group by" operation on input elements of type T, grouping elements according to a classification function, and then performing a reduction operation on the values associated with a given key using the specified downstream Collector.					
static Collector<CharSequence,?,String>				joining()				Returns a Collector that concatenates the input elements into a String, in encounter order.					
static Collector<CharSequence,?,String>				joining(CharSequence delimiter)				Returns a Collector that concatenates the input elements, separated by the specified delimiter, in encounter order.					
static Collector<CharSequence,?,String>				joining(CharSequence delimiter, CharSequence prefix, CharSequence suffix)				Returns a Collector that concatenates the input elements, separated by the specified delimiter, with the specified prefix and suffix, in encounter order.					
static <T,U,A,R> Collector<T,?,R>				mapping(Function<? super T,? extends U> mapper, Collector<? super U,A,R> downstream)				Adapts a Collector accepting elements of type U to one accepting elements of type T by applying a mapping function to each input element before accumulation.					
static <T> Collector<T,?,Optional<T>>				maxBy(Comparator<? super T> comparator)				Returns a Collector that produces the maximal element according to a given Comparator, described as an Optional<T>.					
static <T> Collector<T,?,Optional<T>>				minBy(Comparator<? super T> comparator)				Returns a Collector that produces the minimal element according to a given Comparator, described as an Optional<T>.					
static <T> Collector<T,?,Map<Boolean,List<T>>>				partitioningBy(Predicate<? super T> predicate)				Returns a Collector which partitions the input elements according to a Predicate, and organizes them into a Map<Boolean, List<T>>.					
static <T,D,A> Collector<T,?,Map<Boolean,D>>				partitioningBy(Predicate<? super T> predicate, Collector<? super T,A,D> downstream)				Returns a Collector which partitions the input elements according to a Predicate, reduces the values in each partition according to another Collector, and organizes them into a Map<Boolean, D> whose values are the result of the downstream reduction.					
static <T> Collector<T,?,Optional<T>>				reducing(BinaryOperator<T> op)				Returns a Collector which performs a reduction of its input elements under a specified BinaryOperator.					
static <T> Collector<T,?,T>				reducing(T identity, BinaryOperator<T> op)				Returns a Collector which performs a reduction of its input elements under a					

PREV CLASS	NEXT CLASS	FRAMES	NO FRAMES	ALL CLASSES	SEARCH:
SUMMARY: NESTED FIELD CONSTR METHODDETAIL: FIELD CONSTR METHOD					
				BinaryOperator <U> op)	input elements under a specified mapping function and BinaryOperator .
static <T> Collector <T,?, DoubleSummaryStatistics >				summarizingDouble (ToDoubleFunction <? super T> mapper)	Returns a Collector which applies an double-producing mapping function to each input element, and returns summary statistics for the resulting values.
static <T> Collector <T,?, IntSummaryStatistics >				summarizingInt (ToIntFunction <? super T> mapper)	Returns a Collector which applies an int-producing mapping function to each input element, and returns summary statistics for the resulting values.
static <T> Collector <T,?, LongSummaryStatistics >				summarizingLong (ToLongFunction <? super T> mapper)	Returns a Collector which applies an long-producing mapping function to each input element, and returns summary statistics for the resulting values.
static <T> Collector <T,?, Double >				summingDouble (ToDoubleFunction <? super T> mapper)	Returns a Collector that produces the sum of a double-valued function applied to the input elements.
static <T> Collector <T,?, Integer >				summingInt (ToIntFunction <? super T> mapper)	Returns a Collector that produces the sum of a integer-valued function applied to the input elements.
static <T> Collector <T,?, Long >				summingLong (ToLongFunction <? super T> mapper)	Returns a Collector that produces the sum of a long-valued function applied to the input elements.
static <T,C extends Collection <T>> Collector <T,?,C>				toCollection (Supplier <C> collectionFactory)	Returns a Collector that accumulates the input elements into a new Collection , in encounter order.
static <T,K,U> Collector <T,?, ConcurrentMap <K,U>>				toConcurrentMap (Function <? super T,? extends K> keyMapper, Function <? super T,? extends U> valueMapper)	Returns a concurrent Collector that accumulates elements into a ConcurrentMap whose keys and values are the result of applying the provided mapping functions to the input elements.
static <T,K,U> Collector <T,?, ConcurrentMap <K,U>>				toConcurrentMap (Function <? super T,? extends K> keyMapper, Function <? super T,? extends U> valueMapper, BinaryOperator <U> mergeFunction)	Returns a concurrent Collector that accumulates elements into a ConcurrentMap whose keys and values are the result of applying the provided mapping functions to the input elements.
static <T,K,U,M extends ConcurrentMap <K,U>> Collector <T,?,M>				toConcurrentMap (Function <? super T,? extends K> keyMapper, Function <? super T,? extends U> valueMapper, BinaryOperator <U> mergeFunction, Supplier <M> mapFactory)	Returns a concurrent Collector that accumulates elements into a ConcurrentMap whose keys and values are the result of applying the provided mapping functions to the input elements.
static <T> Collector <T,?, List <T>>				toList ()	Returns a Collector that accumulates the input elements into a new List .

OVERVIEWMODULEPACKAGECLASSUSETREEDEPRECATEDINDEXHELP

PREV CLASSNEXT CLASSFRAMESNO FRAMESALL CLASSES

SUMMARY: NESTED | FIELD | CONSTR | METHODDETAIL: FIELD | CONSTR | METHOD

Java SE 10 & JDK 10

SEARCH:

static <T,K,U> Collector<T,?,Map<K,U>>

toMap(Function<? super T,? extends K> keyMapper, Function<? super T,? extends U> valueMapper, BinaryOperator<U> mergeFunction)

static <T,K,U,M extends Map<K,U>> Collector<T,?,M>

toMap(Function<? super T,? extends K> keyMapper, Function<? super T,? extends U> valueMapper, BinaryOperator<U> mergeFunction, Supplier<M> mapFactory)

static <T> Collector<T,?,Set<T>>

toSet()

static <T> Collector<T,?,List<T>>

toUnmodifiableList()

static <T,K,U> Collector<T,?,Map<K,U>>

toUnmodifiableMap(Function<? super T,? extends K> keyMapper, Function<? super T,? extends U> valueMapper)

static <T,K,U> Collector<T,?,Map<K,U>>

toUnmodifiableMap(Function<? super T,? extends K> keyMapper, Function<? super T,? extends U> valueMapper, BinaryOperator<U> mergeFunction)

static <T> Collector<T,?,Set<T>>

toUnmodifiableSet()

provided mapping functions to the input elements.

Returns a Collector that accumulates elements into a Map whose keys and values are the result of applying the provided mapping functions to the input elements.

Returns a Collector that accumulates the input elements into a new Set.

Returns a Collector that accumulates the input elements into an unmodifiable List in encounter order.

Returns a Collector that accumulates the input elements into an unmodifiable Map, whose keys and values are the result of applying the provided mapping functions to the input elements.

Returns a Collector that accumulates the input elements into an unmodifiable Map, whose keys and values are the result of applying the provided mapping functions to the input elements.

Returns a Collector that accumulates the input elements into an unmodifiable Set.

Methods declared in class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Method Detail

toCollection

public static <T,C extends Collection<T>> Collector<T,?,C> toCollection(Supplier<C> collectionFactory)

Returns a Collector that accumulates the input elements into a new Collection, in encounter order. The Collection is created by the provided factory.

Type Parameters:
T - the type of the input elements
C - the type of the resulting Collection

Parameters:
collectionFactory - a supplier providing a new empty Collection into which the results will be inserted

Returns:

toList

```
public static <T> Collector<T,?,List<T>> toList()
```

Returns a `Collector` that accumulates the input elements into a new `List`. There are no guarantees on the type, mutability, serializability, or thread-safety of the `List` returned; if more control over the returned `List` is required, use `toCollection(Supplier)`.

Type Parameters:

T - the type of the input elements

Returns:

a `Collector` which collects all the input elements into a `List`, in encounter order

toUnmodifiableList

```
public static <T> Collector<T,?,List<T>> toUnmodifiableList()
```

Returns a `Collector` that accumulates the input elements into an `unmodifiable List` in encounter order. The returned `Collector` disallows null values and will throw `NullPointerException` if it is presented with a null value.

Type Parameters:

T - the type of the input elements

Returns:

a `Collector` that accumulates the input elements into an `unmodifiable List` in encounter order

Since:

10

toSet

```
public static <T> Collector<T,?,Set<T>> toSet()
```

Returns a `Collector` that accumulates the input elements into a new `Set`. There are no guarantees on the type, mutability, serializability, or thread-safety of the `Set` returned; if more control over the returned `Set` is required, use `toCollection(Supplier)`.

This is an `unordered` `Collector`.

Type Parameters:

T - the type of the input elements

Returns:

a `Collector` which collects all the input elements into a `Set`

toUnmodifiableSet

```
public static <T> Collector<T,?,Set<T>> toUnmodifiableSet()
```

Returns a `Collector` that accumulates the input elements into an `unmodifiable Set`. The returned `Collector` disallows null values and will throw `NullPointerException` if it is presented with a null value. If the input contains duplicate elements, an arbitrary element of the duplicates is preserved.

This is an `unordered` `Collector`.

Type Parameters:

T - the type of the input elements

Returns:

a `Collector` that accumulates the input elements into an `unmodifiable Set`

Since:

10

joining

a collector that concatenates the input elements into a `String`, in encounter order

joining

```
public static Collector<CharSequence,?,String> joining(CharSequence delimiter)
```

Returns a `Collector` that concatenates the input elements, separated by the specified delimiter, in encounter order.

Parameters:

`delimiter` - the delimiter to be used between each element

Returns:

A `Collector` which concatenates `CharSequence` elements, separated by the specified delimiter, in encounter order

joining

```
public static Collector<CharSequence,?,String> joining(CharSequence delimiter,
                                                         CharSequence prefix,
                                                         CharSequence suffix)
```

Returns a `Collector` that concatenates the input elements, separated by the specified delimiter, with the specified prefix and suffix, in encounter order.

Parameters:

`delimiter` - the delimiter to be used between each element

`prefix` - the sequence of characters to be used at the beginning of the joined result

`suffix` - the sequence of characters to be used at the end of the joined result

Returns:

A `Collector` which concatenates `CharSequence` elements, separated by the specified delimiter, in encounter order

mapping

```
public static <T,U,A,R> Collector<T,?,R> mapping(Function<? super T,? extends U> mapper,
                                                  Collector<? super U,A,R> downstream)
```

Adapts a `Collector` accepting elements of type `U` to one accepting elements of type `T` by applying a mapping function to each input element before accumulation.

API Note:

The `mapping()` collectors are most useful when used in a multi-level reduction, such as downstream of a `groupingBy` or `partitioningBy`. For example, given a stream of `Person`, to accumulate the set of last names in each city:

```
Map<City, Set<String>> lastNamesByCity
= people.stream().collect(
    groupingBy(Person::getCity,
               mapping(Person::getLastName,
                       toSet())));
```

Type Parameters:

`T` - the type of the input elements

`U` - type of elements accepted by downstream collector

`A` - intermediate accumulation type of the downstream collector

`R` - result type of collector

Parameters:

`mapper` - a function to be applied to the input elements

`downstream` - a collector which will accept mapped values

Returns:

a collector which applies the mapping function to the input elements and provides the mapped results to the downstream collector

Collector<? super U,A,R> downstream)

Adapts a Collector accepting elements of type U to one accepting elements of type T by applying a flat mapping function to each input element before accumulation. The flat mapping function maps an input element to a stream covering zero or more output elements that are then accumulated downstream. Each mapped stream is closed after its contents have been placed downstream. (If a mapped stream is null an empty stream is used, instead.)

API Note:
The flatMapping() collectors are most useful when used in a multi-level reduction, such as downstream of a groupingBy or partitioningBy. For example, given a stream of Order, to accumulate the set of line items for each customer:

```
Map<String, Set<LineItem>> itemsByCustomerName
= orders.stream().collect(
    groupingBy(Order::getCustomerName,
        flatMapping(order -> order.getLineItems().stream(),
            toSet())));
```

Type Parameters:
T - the type of the input elements
U - type of elements accepted by downstream collector
A - intermediate accumulation type of the downstream collector
R - result type of collector

Parameters:
mapper - a function to be applied to the input elements, which returns a stream of results
downstream - a collector which will receive the elements of the stream returned by mapper

Returns:
a collector which applies the mapping function to the input elements and provides the flat mapped results to the downstream collector

Since:
9

filtering

```
public static <T,A,R> Collector<T,?,R> filtering(Predicate<? super T> predicate,
    Collector<? super T,A,R> downstream)
```

Adapts a Collector to one accepting elements of the same type T by applying the predicate to each input element and only accumulating if the predicate returns true.

API Note:
The filtering() collectors are most useful when used in a multi-level reduction, such as downstream of a groupingBy or partitioningBy. For example, given a stream of Employee, to accumulate the employees in each department that have a salary above a certain threshold:

```
Map<Department, Set<Employee>> wellPaidEmployeesByDepartment
= employees.stream().collect(
    groupingBy(Employee::getDepartment,
        filtering(e -> e.getSalary() > 2000,
            toSet())));
```

A filtering collector differs from a stream's filter() operation. In this example, suppose there are no employees whose salary is above the threshold in some department. Using a filtering collector as shown above would result in a mapping from that department to an empty Set. If a stream filter() operation were done instead, there would be no mapping for that department at all.

Type Parameters:
T - the type of the input elements
A - intermediate accumulation type of the downstream collector
R - result type of collector

Parameters:
predicate - a predicate to be applied to the input elements
downstream - a collector which will accept values that match the predicate

collectingAndThen

```
public static <T,A,R,RR> Collector<T,A,RR> collectingAndThen(Collector<T,A,R> downstream,
                                                            Function<R,RR> finisher)
```

Adapts a `Collector` to perform an additional finishing transformation. For example, one could adapt the `toList()` collector to always produce an immutable list with:

```
List<String> list = people.stream().collect(
    collectingAndThen(toList(),
        Collections::unmodifiableList));
```

Type Parameters:

T - the type of the input elements

A - intermediate accumulation type of the downstream collector

R - result type of the downstream collector

RR - result type of the resulting collector

Parameters:

downstream - a collector

finisher - a function to be applied to the final result of the downstream collector

Returns:

a collector which performs the action of the downstream collector, followed by an additional finishing step

counting

```
public static <T> Collector<T,?,Long> counting()
```

Returns a `Collector` accepting elements of type T that counts the number of input elements. If no elements are present, the result is 0.

Implementation Requirements:

This produces a result equivalent to:

```
reducing(0L, e -> 1L, Long::sum)
```

Type Parameters:

T - the type of the input elements

Returns:

a `Collector` that counts the input elements

minBy

```
public static <T> Collector<T,?,Optional<T>> minBy(Comparator<? super T> comparator)
```

Returns a `Collector` that produces the minimal element according to a given `Comparator`, described as an `Optional<T>`.

Implementation Requirements:

This produces a result equivalent to:

```
reducing(BinaryOperator.minBy(comparator))
```

Type Parameters:

T - the type of the input elements

Parameters:

comparator - a `Comparator` for comparing elements

maxBy

```
public static <T> Collector<T,?,Optional<T>> maxBy(Comparator<? super T> comparator)
```

Returns a Collector that produces the maximal element according to a given Comparator, described as an Optional<T>.

Implementation Requirements:

This produces a result equivalent to:

```
reducing(BinaryOperator.maxBy(comparator))
```

Type Parameters:

T - the type of the input elements

Parameters:

comparator - a Comparator for comparing elements

Returns:

a Collector that produces the maximal value

summingInt

```
public static <T> Collector<T,?,Integer> summingInt(ToIntFunction<? super T> mapper)
```

Returns a Collector that produces the sum of a integer-valued function applied to the input elements. If no elements are present, the result is 0.

Type Parameters:

T - the type of the input elements

Parameters:

mapper - a function extracting the property to be summed

Returns:

a Collector that produces the sum of a derived property

summingLong

```
public static <T> Collector<T,?,Long> summingLong(ToLongFunction<? super T> mapper)
```

Returns a Collector that produces the sum of a long-valued function applied to the input elements. If no elements are present, the result is 0.

Type Parameters:

T - the type of the input elements

Parameters:

mapper - a function extracting the property to be summed

Returns:

a Collector that produces the sum of a derived property

summingDouble

```
public static <T> Collector<T,?,Double> summingDouble(ToDoubleFunction<? super T> mapper)
```

Returns a Collector that produces the sum of a double-valued function applied to the input elements. If no elements are present, the result is 0.

The sum returned can vary depending upon the order in which values are recorded, due to accumulated rounding error in addition of values of differing magnitudes. Values sorted by increasing absolute magnitude tend to yield more accurate results. If any recorded value is a NaN or the sum is at any point a NaN then the sum will be NaN.

Type Parameters:

T - the type of the input elements

Parameters:

averagingInt

```
public static <T> Collector<T,?,Double> averagingInt(ToIntFunction<? super T> mapper)
```

Returns a Collector that produces the arithmetic mean of an integer-valued function applied to the input elements. If no elements are present, the result is 0.

Type Parameters:

T - the type of the input elements

Parameters:

mapper - a function extracting the property to be averaged

Returns:

a Collector that produces the arithmetic mean of a derived property

averagingLong

```
public static <T> Collector<T,?,Double> averagingLong(ToLongFunction<? super T> mapper)
```

Returns a Collector that produces the arithmetic mean of a long-valued function applied to the input elements. If no elements are present, the result is 0.

Type Parameters:

T - the type of the input elements

Parameters:

mapper - a function extracting the property to be averaged

Returns:

a Collector that produces the arithmetic mean of a derived property

averagingDouble

```
public static <T> Collector<T,?,Double> averagingDouble(ToDoubleFunction<? super T> mapper)
```

Returns a Collector that produces the arithmetic mean of a double-valued function applied to the input elements. If no elements are present, the result is 0.

The average returned can vary depending upon the order in which values are recorded, due to accumulated rounding error in addition of values of differing magnitudes. Values sorted by increasing absolute magnitude tend to yield more accurate results. If any recorded value is a NaN or the sum is at any point a NaN then the average will be NaN.

Implementation Note:

The double format can represent all consecutive integers in the range -2^{53} to 2^{53} . If the pipeline has more than 2^{53} values, the divisor in the average computation will saturate at 2^{53} , leading to additional numerical errors.

Type Parameters:

T - the type of the input elements

Parameters:

mapper - a function extracting the property to be averaged

Returns:

a Collector that produces the arithmetic mean of a derived property

reducing

```
public static <T> Collector<T,?,T> reducing(T identity,  
                                           BinaryOperator<T> op)
```

Returns a Collector which performs a reduction of its input elements under a specified BinaryOperator using the provided identity.

API Note:

Parameters:
identity - the identity value for the reduction (also, the value that is returned when there are no input elements)
op - a BinaryOperator<T> used to reduce the input elements

Returns:
a Collector which implements the reduction operation

See Also:
reducing(BinaryOperator), reducing(Object, Function, BinaryOperator)

reducing

```
public static <T> Collector<T,?,Optional<T>> reducing(BinaryOperator<T> op)
```

Returns a Collector which performs a reduction of its input elements under a specified BinaryOperator. The result is described as an Optional<T>.

API Note:
The reducing() collectors are most useful when used in a multi-level reduction, downstream of groupingBy or partitioningBy. To perform a simple reduction on a stream, use Stream.reduce(BinaryOperator) instead.

For example, given a stream of Person, to calculate tallest person in each city:

```
Comparator<Person> byHeight = Comparator.comparing(Person::getHeight);
Map<City, Optional<Person>> tallestByCity
    = people.stream().collect(
        groupingBy(Person::getCity,
            reducing(BinaryOperator.maxBy(byHeight))));
```

Type Parameters:
T - element type for the input and output of the reduction

Parameters:
op - a BinaryOperator<T> used to reduce the input elements

Returns:
a Collector which implements the reduction operation

See Also:
reducing(Object, BinaryOperator), reducing(Object, Function, BinaryOperator)

reducing

```
public static <T,U> Collector<T,?,U> reducing(U identity,
                                             Function<? super T,? extends U> mapper,
                                             BinaryOperator<U> op)
```

Returns a Collector which performs a reduction of its input elements under a specified mapping function and BinaryOperator. This is a generalization of reducing(Object, BinaryOperator) which allows a transformation of the elements before reduction.

API Note:
The reducing() collectors are most useful when used in a multi-level reduction, downstream of groupingBy or partitioningBy. To perform a simple map-reduce on a stream, use Stream.map(Function) and Stream.reduce(Object, BinaryOperator) instead.

For example, given a stream of Person, to calculate the longest last name of residents in each city:

```
Comparator<String> byLength = Comparator.comparing(String::length);
Map<City, String> longestLastNameByCity
    = people.stream().collect(
        groupingBy(Person::getCity,
            reducing("",
                Person::getLastName,
                BinaryOperator.maxBy(byLength))));
```

Type Parameters:

PREV CLASSNEXT CLASSFRAMESNO FRAMESALL CLASSES

SEARCH:

SUMMARY: NESTED | FIELD | CONSTR | METHODDETAIL: FIELD | CONSTR | METHOD

identity - the identity value for the reduction (also, the value that is returned when there are no input elements),

mapper - a mapping function to apply to each input value

op - a BinaryOperator<U> used to reduce the mapped values

Returns:

a Collector implementing the map-reduce operation

See Also:

reducing(Object, BinaryOperator), reducing(BinaryOperator)

groupingBy

public static <T,K> Collector<T,?,Map<K,List<T>>> groupingBy(Function<? super T,? extends K> classifier)

Returns a Collector implementing a "group by" operation on input elements of type T, grouping elements according to a classification function, and returning the results in a Map.

The classification function maps elements to some key type K. The collector produces a Map<K, List<T>> whose keys are the values resulting from applying the classification function to the input elements, and whose corresponding values are Lists containing the input elements which map to the associated key under the classification function.

There are no guarantees on the type, mutability, serializability, or thread-safety of the Map or List objects returned.

Implementation Requirements:

This produces a result similar to:

```
groupingBy(classifier, toList());
```

Implementation Note:

The returned Collector is not concurrent. For parallel stream pipelines, the combiner function operates by merging the keys from one map into another, which can be an expensive operation. If preservation of the order in which elements appear in the resulting Map collector is not required, using groupingByConcurrent(Function) may offer better parallel performance.

Type Parameters:

T - the type of the input elements

K - the type of the keys

Parameters:

classifier - the classifier function mapping input elements to keys

Returns:

a Collector implementing the group-by operation

See Also:

groupingBy(Function, Collector), groupingBy(Function, Supplier, Collector), groupingByConcurrent(Function)

groupingBy

public static <T,K,A,D> Collector<T,?,Map<K,D>> groupingBy(Function<? super T,? extends K> classifier,
Collector<? super T,A,D> downstream)

Returns a Collector implementing a cascaded "group by" operation on input elements of type T, grouping elements according to a classification function, and then performing a reduction operation on the values associated with a given key using the specified downstream Collector.

The classification function maps elements to some key type K. The downstream collector operates on elements of type T and produces a result of type D. The resulting collector produces a Map<K, D>.

There are no guarantees on the type, mutability, serializability, or thread-safety of the Map returned.

For example, to compute the set of last names of people in each city:

```
Map<City, Set<String>> namesByCity  
= people.stream().collect(  
    groupingBy(Person::getCity,  
        mapping(Person::getLastName,
```

into another, which can be an expensive operation. If preservation of the order in which elements are presented to the downstream collector is not required, using `groupingByConcurrent(Function, Collector)` may offer better parallel performance.

Type Parameters:

T - the type of the input elements

K - the type of the keys

A - the intermediate accumulation type of the downstream collector

D - the result type of the downstream reduction

Parameters:

classifier - a classifier function mapping input elements to keys

downstream - a Collector implementing the downstream reduction

Returns:

a Collector implementing the cascaded group-by operation

See Also:

`groupingBy(Function)`, `groupingBy(Function, Supplier, Collector)`, `groupingByConcurrent(Function, Collector)`

groupingBy

```
public static <T,K,D,A,M extends Map<K,D>> Collector<T,?,M> groupingBy(Function<? super T,? extends K> classifier,
                                                                    Supplier<M> mapFactory,
                                                                    Collector<? super T,A,D> downstream)
```

Returns a Collector implementing a cascaded "group by" operation on input elements of type T, grouping elements according to a classification function, and then performing a reduction operation on the values associated with a given key using the specified downstream Collector. The Map produced by the Collector is created with the supplied factory function.

The classification function maps elements to some key type K. The downstream collector operates on elements of type T and produces a result of type D. The resulting collector produces a Map<K, D>.

For example, to compute the set of last names of people in each city, where the city names are sorted:

```
Map<City, Set<String>> namesByCity
= people.stream().collect(
    groupingBy(Person::getCity,
               TreeMap::new,
               mapping(Person::getLastName,
                       toSet())));
```

Implementation Note:

The returned Collector is not concurrent. For parallel stream pipelines, the combiner function operates by merging the keys from one map into another, which can be an expensive operation. If preservation of the order in which elements are presented to the downstream collector is not required, using `groupingByConcurrent(Function, Supplier, Collector)` may offer better parallel performance.

Type Parameters:

T - the type of the input elements

K - the type of the keys

A - the intermediate accumulation type of the downstream collector

D - the result type of the downstream reduction

M - the type of the resulting Map

Parameters:

classifier - a classifier function mapping input elements to keys

downstream - a Collector implementing the downstream reduction

mapFactory - a supplier providing a new empty Map into which the results will be inserted

Returns:

a Collector implementing the cascaded group-by operation

See Also:

`groupingBy(Function, Collector)`, `groupingBy(Function)`, `groupingByConcurrent(Function, Supplier, Collector)`

Returns a concurrent Collector implementing a "group by" operation on input elements of type T, grouping elements according to a classification function.

This is a `concurrent` and `unordered` Collector.

The classification function maps elements to some key type K. The collector produces a `ConcurrentMap<K, List<T>>` whose keys are the values resulting from applying the classification function to the input elements, and whose corresponding values are `List`s containing the input elements which map to the associated key under the classification function.

There are no guarantees on the type, mutability, or serializability of the `ConcurrentMap` or `List` objects returned, or of the thread-safety of the `List` objects returned.

Implementation Requirements:

This produces a result similar to:

```
groupingByConcurrent(classifier, toList());
```

Type Parameters:

T - the type of the input elements

K - the type of the keys

Parameters:

classifier - a classifier function mapping input elements to keys

Returns:

a concurrent, unordered Collector implementing the group-by operation

See Also:

`groupingBy(Function)`, `groupingByConcurrent(Function, Collector)`, `groupingByConcurrent(Function, Supplier, Collector)`

groupingByConcurrent

```
public static <T,K,A,D> Collector<T,?,ConcurrentMap<K,D>> groupingByConcurrent(Function<? super T,? extends K> classifier,
                                                                              Collector<? super T,A,D> downstream)
```

Returns a concurrent Collector implementing a cascaded "group by" operation on input elements of type T, grouping elements according to a classification function, and then performing a reduction operation on the values associated with a given key using the specified downstream Collector.

This is a `concurrent` and `unordered` Collector.

The classification function maps elements to some key type K. The downstream collector operates on elements of type T and produces a result of type D. The resulting collector produces a `ConcurrentMap<K, D>`.

There are no guarantees on the type, mutability, or serializability of the `ConcurrentMap` returned.

For example, to compute the set of last names of people in each city, where the city names are sorted:

```
ConcurrentMap<City, Set<String>> namesByCity
= people.stream().collect(
    groupingByConcurrent(Person::getCity,
                        mapping(Person::getLastName,
                                toSet())));
```

Type Parameters:

T - the type of the input elements

K - the type of the keys

A - the intermediate accumulation type of the downstream collector

D - the result type of the downstream reduction

Parameters:

classifier - a classifier function mapping input elements to keys

downstream - a Collector implementing the downstream reduction

Returns:

groupingByConcurrent

```
public static <T,K,A,D,M extends ConcurrentMap<K,D>> Collector<T,?,M> groupingByConcurrent(Function<? super T,? extends K> clas:
                                                                    Supplier<M> mapFactory,
                                                                    Collector<? super T,A,D> downstream)
```

Returns a concurrent Collector implementing a cascaded "group by" operation on input elements of type T, grouping elements according to a classification function, and then performing a reduction operation on the values associated with a given key using the specified downstream Collector. The ConcurrentMap produced by the Collector is created with the supplied factory function.

This is a concurrent and unordered Collector.

The classification function maps elements to some key type K. The downstream collector operates on elements of type T and produces a result of type D. The resulting collector produces a ConcurrentMap<K, D>.

For example, to compute the set of last names of people in each city, where the city names are sorted:

```
ConcurrentMap<City, Set<String>> namesByCity
= people.stream().collect(
    groupingByConcurrent(Person::getCity,
                          ConcurrentSkipListMap::new,
                          mapping(Person::getLastName,
                                toSet())));
```

Type Parameters:

- T - the type of the input elements
- K - the type of the keys
- A - the intermediate accumulation type of the downstream collector
- D - the result type of the downstream reduction
- M - the type of the resulting ConcurrentMap

Parameters:

- classifier - a classifier function mapping input elements to keys
- downstream - a Collector implementing the downstream reduction
- mapFactory - a supplier providing a new empty ConcurrentMap into which the results will be inserted

Returns:

a concurrent, unordered Collector implementing the cascaded group-by operation

See Also:

groupingByConcurrent(Function), groupingByConcurrent(Function, Collector), groupingBy(Function, Supplier, Collector)

partitioningBy

```
public static <T> Collector<T,?,Map<Boolean,List<T>>> partitioningBy(Predicate<? super T> predicate)
```

Returns a Collector which partitions the input elements according to a Predicate, and organizes them into a Map<Boolean, List<T>>. The returned Map always contains mappings for both false and true keys. There are no guarantees on the type, mutability, serializability, or thread-safety of the Map or List returned.

API Note:

If a partition has no elements, its value in the result Map will be an empty List.

Type Parameters:

- T - the type of the input elements

Parameters:

- predicate - a predicate used for classifying input elements

Returns:

a Collector implementing the partitioning operation

See Also:

partitioningBy(Predicate, Collector)

PREV CLASSNEXT CLASSFRAMESNO FRAMESALL CLASSES

SEARCH:

SUMMARY: NESTED | FIELD | CONSTR | METHOD

DETAIL: FIELD | CONSTR | METHOD

Collector<? super T,A,D> downstream)

Returns a `Collector` which partitions the input elements according to a `Predicate`, reduces the values in each partition according to another `Collector`, and organizes them into a `Map<Boolean, D>` whose values are the result of the downstream reduction.

The returned `Map` always contains mappings for both `false` and `true` keys. There are no guarantees on the type, mutability, serializability, or thread-safety of the `Map` returned.

API Note:

If a partition has no elements, its value in the result `Map` will be obtained by calling the downstream collector's supplier function and then applying the finisher function.

Type Parameters:

- T - the type of the input elements
- A - the intermediate accumulation type of the downstream collector
- D - the result type of the downstream reduction

Parameters:

- predicate - a predicate used for classifying input elements
- downstream - a `Collector` implementing the downstream reduction

Returns:

a `Collector` implementing the cascaded partitioning operation

See Also:

`partitioningBy(Predicate)`

toMap

```
public static <T,K,U> Collector<T,?,Map<K,U>> toMap(Function<? super T,? extends K> keyMapper,
                                                    Function<? super T,? extends U> valueMapper)
```

Returns a `Collector` that accumulates elements into a `Map` whose keys and values are the result of applying the provided mapping functions to the input elements.

If the mapped keys contain duplicates (according to `Object.equals(Object)`), an `IllegalStateException` is thrown when the collection operation is performed. If the mapped keys might have duplicates, use `toMap(Function, Function, BinaryOperator)` instead.

There are no guarantees on the type, mutability, serializability, or thread-safety of the `Map` returned.

API Note:

It is common for either the key or the value to be the input elements. In this case, the utility method `Function.identity()` may be helpful. For example, the following produces a `Map` mapping students to their grade point average:

```
Map<Student, Double> studentToGPA
= students.stream().collect(
    toMap(Function.identity(),
          student -> computeGPA(student)));
```

And the following produces a `Map` mapping a unique identifier to students:

```
Map<String, Student> studentIdToStudent
= students.stream().collect(
    toMap(Student::getId,
          Function.identity()));
```

Implementation Note:

The returned `Collector` is not concurrent. For parallel stream pipelines, the combiner function operates by merging the keys from one map into another, which can be an expensive operation. If it is not required that results are inserted into the `Map` in encounter order, using `toConcurrentMap(Function, Function)` may offer better parallel performance.

Type Parameters:

- T - the type of the input elements
- K - the output type of the key mapping function
- U - the output type of the value mapping function

a Collector which collects elements into a Map whose keys and values are the result of applying mapping functions to the input elements

See Also:

`toMap(Function, Function, BinaryOperator)`, `toMap(Function, Function, BinaryOperator, Supplier)`, `toConcurrentMap(Function, Function)`

toUnmodifiableMap

```
public static <T,K,U> Collector<T,?,Map<K,U>> toUnmodifiableMap(Function<? super T,? extends K> keyMapper,
                                                                Function<? super T,? extends U> valueMapper)
```

Returns a Collector that accumulates the input elements into an [unmodifiable Map](#), whose keys and values are the result of applying the provided mapping functions to the input elements.

If the mapped keys contain duplicates (according to `Object.equals(Object)`), an `IllegalStateException` is thrown when the collection operation is performed. If the mapped keys might have duplicates, use `toUnmodifiableMap(Function, Function, BinaryOperator)` to handle merging of the values.

The returned Collector disallows null keys and values. If either mapping function returns null, `NullPointerException` will be thrown.

Type Parameters:

T - the type of the input elements

K - the output type of the key mapping function

U - the output type of the value mapping function

Parameters:

keyMapper - a mapping function to produce keys, must be non-null

valueMapper - a mapping function to produce values, must be non-null

Returns:

a Collector that accumulates the input elements into an [unmodifiable Map](#), whose keys and values are the result of applying the provided mapping functions to the input elements

Throws:

[NullPointerException](#) - if either keyMapper or valueMapper is null

Since:

10

See Also:

`toUnmodifiableMap(Function, Function, BinaryOperator)`

toMap

```
public static <T,K,U> Collector<T,?,Map<K,U>> toMap(Function<? super T,? extends K> keyMapper,
                                                    Function<? super T,? extends U> valueMapper,
                                                    BinaryOperator<U> mergeFunction)
```

Returns a Collector that accumulates elements into a Map whose keys and values are the result of applying the provided mapping functions to the input elements.

If the mapped keys contain duplicates (according to `Object.equals(Object)`), the value mapping function is applied to each equal element, and the results are merged using the provided merging function.

There are no guarantees on the type, mutability, serializability, or thread-safety of the Map returned.

API Note:

There are multiple ways to deal with collisions between multiple elements mapping to the same key. The other forms of `toMap` simply use a merge function that throws unconditionally, but you can easily write more flexible merge policies. For example, if you have a stream of `Person`, and you want to produce a "phone book" mapping name to address, but it is possible that two persons have the same name, you can do as follows to gracefully deal with these collisions, and produce a Map mapping names to a concatenated list of addresses:

```
Map<String, String> phoneBook
= people.stream().collect(
    toMap(Person::getName,
          Person::getAddress,
```

SUMMARY: NESTED | FIELD | CONSTR | METHODDETAIL: FIELD | CONSTR | METHOD

into another, which can be an expensive operation. If it is not required that results are merged into the Map in encounter order, using `toConcurrentMap(Function, Function, BinaryOperator)` may offer better parallel performance.

Type Parameters:

T - the type of the input elements

K - the output type of the key mapping function

U - the output type of the value mapping function

Parameters:

keyMapper - a mapping function to produce keys

valueMapper - a mapping function to produce values

mergeFunction - a merge function, used to resolve collisions between values associated with the same key, as supplied to `Map.merge(Object, Object, BiFunction)`

Returns:

a Collector which collects elements into a Map whose keys are the result of applying a key mapping function to the input elements, and whose values are the result of applying a value mapping function to all input elements equal to the key and combining them using the merge function

See Also:

`toMap(Function, Function)`, `toMap(Function, Function, BinaryOperator, Supplier)`, `toConcurrentMap(Function, Function, BinaryOperator)`

toUnmodifiableMap

```
public static <T,K,U> Collector<T,?,Map<K,U>> toUnmodifiableMap(Function<? super T,? extends K> keyMapper,
                                                                Function<? super T,? extends U> valueMapper,
                                                                BinaryOperator<U> mergeFunction)
```

Returns a Collector that accumulates the input elements into an `unmodifiable Map`, whose keys and values are the result of applying the provided mapping functions to the input elements.

If the mapped keys contain duplicates (according to `Object.equals(Object)`), the value mapping function is applied to each equal element, and the results are merged using the provided merging function.

The returned Collector disallows null keys and values. If either mapping function returns null, `NullPointerException` will be thrown.

Type Parameters:

T - the type of the input elements

K - the output type of the key mapping function

U - the output type of the value mapping function

Parameters:

keyMapper - a mapping function to produce keys, must be non-null

valueMapper - a mapping function to produce values, must be non-null

mergeFunction - a merge function, used to resolve collisions between values associated with the same key, as supplied to `Map.merge(Object, Object, BiFunction)`, must be non-null

Returns:

a Collector that accumulates the input elements into an `unmodifiable Map`, whose keys and values are the result of applying the provided mapping functions to the input elements

Throws:

`NullPointerException` - if the keyMapper, valueMapper, or mergeFunction is null

Since:

10

See Also:

`toUnmodifiableMap(Function, Function)`

toMap

```
public static <T,K,U,M extends Map<K,U>> Collector<T,?,M> toMap(Function<? super T,? extends K> keyMapper,
                                                                Function<? super T,? extends U> valueMapper,
```

If the mapped keys contain duplicates (according to `Object.equals(Object)`), the value mapping function is applied to each equal element, and the results are merged using the provided merging function. The Map is created by a provided supplier function.

Implementation Note:

The returned Collector is not concurrent. For parallel stream pipelines, the combiner function operates by merging the keys from one map into another, which can be an expensive operation. If it is not required that results are merged into the Map in encounter order, using `toConcurrentMap(Function, Function, BinaryOperator, Supplier)` may offer better parallel performance.

Type Parameters:

T - the type of the input elements

K - the output type of the key mapping function

U - the output type of the value mapping function

M - the type of the resulting Map

Parameters:

keyMapper - a mapping function to produce keys

valueMapper - a mapping function to produce values

mergeFunction - a merge function, used to resolve collisions between values associated with the same key, as supplied to `Map.merge(Object, Object, BiFunction)`

mapFactory - a supplier providing a new empty Map into which the results will be inserted

Returns:

a Collector which collects elements into a Map whose keys are the result of applying a key mapping function to the input elements, and whose values are the result of applying a value mapping function to all input elements equal to the key and combining them using the merge function

See Also:

`toMap(Function, Function)`, `toMap(Function, Function, BinaryOperator)`, `toConcurrentMap(Function, Function, BinaryOperator, Supplier)`

toConcurrentMap

```
public static <T,K,U> Collector<T,?,ConcurrentMap<K,U>> toConcurrentMap(Function<? super T,? extends K> keyMapper,
                                                                    Function<? super T,? extends U> valueMapper)
```

Returns a concurrent Collector that accumulates elements into a ConcurrentMap whose keys and values are the result of applying the provided mapping functions to the input elements.

If the mapped keys contain duplicates (according to `Object.equals(Object)`), an `IllegalStateException` is thrown when the collection operation is performed. If the mapped keys may have duplicates, use `toConcurrentMap(Function, Function, BinaryOperator)` instead.

There are no guarantees on the type, mutability, or serializability of the ConcurrentMap returned.

API Note:

It is common for either the key or the value to be the input elements. In this case, the utility method `Function.identity()` may be helpful. For example, the following produces a ConcurrentMap mapping students to their grade point average:

```
ConcurrentMap<Student, Double> studentToGPA
= students.stream().collect(
    toConcurrentMap(Function.identity(),
                    student -> computeGPA(student)));
```

And the following produces a ConcurrentMap mapping a unique identifier to students:

```
ConcurrentMap<String, Student> studentIdToStudent
= students.stream().collect(
    toConcurrentMap(Student::getId,
                    Function.identity()));
```

This is a [concurrent](#) and [unordered](#) Collector.

Type Parameters:

T - the type of the input elements

K - the output type of the key mapping function

SUMMARY: NESTED | FIELD | CONSTR | METHODDETAIL: FIELD | CONSTR | METHOD

`toConcurrentMap` - the mapping function to produce values

Returns:
a concurrent, unordered `Collector` which collects elements into a `ConcurrentMap` whose keys are the result of applying a key mapping function to the input elements, and whose values are the result of applying a value mapping function to the input elements

See Also:
`toMap(Function, Function)`, `toConcurrentMap(Function, Function, BinaryOperator)`, `toConcurrentMap(Function, Function, BinaryOperator, Supplier)`

toConcurrentMap

```
public static <T,K,U> Collector<T,?,ConcurrentMap<K,U>> toConcurrentMap(Function<? super T,? extends K> keyMapper,
                                                                    Function<? super T,? extends U> valueMapper,
                                                                    BinaryOperator<U> mergeFunction)
```

Returns a concurrent `Collector` that accumulates elements into a `ConcurrentMap` whose keys and values are the result of applying the provided mapping functions to the input elements.

If the mapped keys contain duplicates (according to `Object.equals(Object)`), the value mapping function is applied to each equal element, and the results are merged using the provided merging function.

There are no guarantees on the type, mutability, or serializability of the `ConcurrentMap` returned.

API Note:
There are multiple ways to deal with collisions between multiple elements mapping to the same key. The other forms of `toConcurrentMap` simply use a merge function that throws unconditionally, but you can easily write more flexible merge policies. For example, if you have a stream of `Person`, and you want to produce a "phone book" mapping name to address, but it is possible that two persons have the same name, you can do as follows to gracefully deal with these collisions, and produce a `ConcurrentMap` mapping names to a concatenated list of addresses:

```
ConcurrentMap<String, String> phoneBook
= people.stream().collect(
    toConcurrentMap(Person::getName,
                    Person::getAddress,
                    (s, a) -> s + ", " + a));
```

This is a `concurrent` and `unordered` `Collector`.

Type Parameters:
T - the type of the input elements
K - the output type of the key mapping function
U - the output type of the value mapping function

Parameters:
`keyMapper` - a mapping function to produce keys
`valueMapper` - a mapping function to produce values
`mergeFunction` - a merge function, used to resolve collisions between values associated with the same key, as supplied to `Map.merge(Object, Object, BiFunction)`

Returns:
a concurrent, unordered `Collector` which collects elements into a `ConcurrentMap` whose keys are the result of applying a key mapping function to the input elements, and whose values are the result of applying a value mapping function to all input elements equal to the key and combining them using the merge function

See Also:
`toConcurrentMap(Function, Function)`, `toConcurrentMap(Function, Function, BinaryOperator, Supplier)`, `toMap(Function, Function, BinaryOperator)`

toConcurrentMap

```
public static <T,K,U,M extends ConcurrentMap<K,U>> Collector<T,?,M> toConcurrentMap(Function<? super T,? extends K> keyMapper,
                                                                    Function<? super T,? extends U> valueMapper,
                                                                    BinaryOperator<U> mergeFunction,
                                                                    Supplier<M> mapFactory)
```

This is a `concurrent` and `unordered` Collector.

Type Parameters:

T - the type of the input elements

K - the output type of the key mapping function

U - the output type of the value mapping function

M - the type of the resulting `ConcurrentMap`

Parameters:

keyMapper - a mapping function to produce keys

valueMapper - a mapping function to produce values

mergeFunction - a merge function, used to resolve collisions between values associated with the same key, as supplied to `Map.merge(Object, Object, BiFunction)`

mapFactory - a supplier providing a new empty `ConcurrentMap` into which the results will be inserted

Returns:

a concurrent, unordered Collector which collects elements into a `ConcurrentMap` whose keys are the result of applying a key mapping function to the input elements, and whose values are the result of applying a value mapping function to all input elements equal to the key and combining them using the merge function

See Also:

`toConcurrentMap(Function, Function)`, `toConcurrentMap(Function, Function, BinaryOperator)`, `toMap(Function, Function, BinaryOperator, Supplier)`

summarizingInt

```
public static <T> Collector<T,?,IntSummaryStatistics> summarizingInt(ToIntFunction<? super T> mapper)
```

Returns a Collector which applies an int-producing mapping function to each input element, and returns summary statistics for the resulting values.

Type Parameters:

T - the type of the input elements

Parameters:

mapper - a mapping function to apply to each element

Returns:

a Collector implementing the summary-statistics reduction

See Also:

`summarizingDouble(ToDoubleFunction)`, `summarizingLong(ToLongFunction)`

summarizingLong

```
public static <T> Collector<T,?,LongSummaryStatistics> summarizingLong(ToLongFunction<? super T> mapper)
```

Returns a Collector which applies an long-producing mapping function to each input element, and returns summary statistics for the resulting values.

Type Parameters:

T - the type of the input elements

Parameters:

mapper - the mapping function to apply to each element

Returns:

a Collector implementing the summary-statistics reduction

See Also:

`summarizingDouble(ToDoubleFunction)`, `summarizingInt(ToIntFunction)`

summarizingDouble

OVERVIEW MODULE PACKAGE CLASS USE TREE DEPRECATED INDEX HELP

Java SE 10 & JDK 10

PREV CLASS NEXT CLASS FRAMES NO FRAMES ALL CLASSES

SEARCH:

SUMMARY: NESTED | FIELD | CONSTR | METHOD DETAIL: FIELD | CONSTR | METHOD

type parameters:

T - the type of the input elements

Parameters:

mapper - a mapping function to apply to each element

Returns:

a Collector implementing the summary-statistics reduction

See Also:

summarizingLong(ToLongFunction), summarizingInt(ToIntFunction)

OVERVIEW MODULE PACKAGE CLASS USE TREE DEPRECATED INDEX HELP

Java SE 10 & JDK 10

PREV CLASS NEXT CLASS FRAMES NO FRAMES ALL CLASSES

SUMMARY: NESTED | FIELD | CONSTR | METHOD DETAIL: FIELD | CONSTR | METHOD

Report a bug or suggest an enhancement

For further API reference and developer documentation see the [Java SE Documentation](#), which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples.

Java is a trademark or registered trademark of Oracle and/or its affiliates in the US and other countries.

Copyright © 1993, 2018, Oracle and/or its affiliates, 500 Oracle Parkway, Redwood Shores, CA 94065 USA.

All rights reserved. Use is subject to [license terms](#) and the [documentation redistribution policy](#). [Modify Cookie Preferences](#). [Modify Ad Choices](#).